

Minghao JIANG | 姜明昊

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EDUCATION

Ph.D. in Bioinformatics at The University of Hong Kong, Hong Kong, Hong Kong SAR Sep 2023 – present

- Department: School of Biomedical Sciences, Li Ka Shing Faculty of Medicine
- Supervisor: Professor Jason Wing Hon WONG

M.Sc. in Biology at Shanghai Jiao Tong University, Shanghai, China Sep 2019 – Jun 2022

- Department: Shanghai Institute of Hematology, State Key Laboratory of Medical Genomics
- Supervisor: Professor Lin CHENG
- GPA: 3.90 / 4.0

B.Sc. in Bioinformatics at Harbin Medical University, Harbin, Heilongjiang Province, China Sep 2014 – Jul 2019

- College: College of Bioinformatics Science and Technology
- Average score: 81.41 / 100 (five-year program)

RESEARCH INTERESTS

Cancer genomics | Leukemia | Multi-omics integration | Genome organization | RNA biology | Bioinformatics | Computational biology

RESEARCH EXPERIENCE

Ph.D. research, School of Biomedical Sciences, The University of Hong Kong Sep 2023 – present

- Mapping multi-omics determinants of relapse and therapeutic resistance in childhood acute lymphoblastic leukemia.
- Dissecting the mechanisms underlying APOBEC-associated DNA and RNA mutational signatures in *ETV6::RUNX1*-subtype B-ALL.
- Developing **gghic**, an R package for exploring and visualizing 3D genome organization from Hi-C and related chromatin-contact data (manuscript in preparation; first-author preprint, *arXiv*, 2024; first-author poster, GIW XXXIV ISCB-Asia 2025).
- Characterizing structural-variant heterogeneity and therapeutic vulnerabilities in acute lymphoblastic leukemia via long-read multi-omics profiling (manuscript in preparation; first-author poster, SBMS RPG Symposium 2025; co-authored abstract, ASH 2025; first-author talk, SBMS RPG Symposium 2026).
- Contributed bioinformatics analyses to a study of CUDC-907 in *MEF2D* fusion-driven BCP-ALL (co-author, *Signal Transduct Target Ther.*, 2025).

M.Sc. research, Shanghai Institute of Hematology, Shanghai Jiao Tong University Sep 2019 – Jun 2022

- Benchmarked 21 differential alternative-splicing tools for RNA-seq at the event level and proposed a tool-selection protocol (first-author, *Brief Bioinform.*, 2023).
- Revealed that TCF12 acts as a co-factor of IGH::DUX4 to enhance transactivation via a positive-feedback loop in B-ALL (co-first author, *Clin Transl Med.*, 2023).
- Modelled the spread of the SARS-CoV-2 Omicron epidemic in Shanghai using a time-delay differential equation framework (co-first author, *Front Med (Lausanne)*, 2022).
- Identified prognosis-associated cell-cell communication in B-cell acute lymphoblastic leukemia (B-ALL) from single-cell transcriptomes (co-first author, *Front Oncol.*, 2021).
- Contributed multi-omics analyses (RNA-seq, ChIP-seq, scRNA-seq, CyTOF) to studies on PML phase separation, MEF2D::HNRNPUL1, and RAG1/2-driven oncogenic splicing.

B.Sc. research, College of Bioinformatics Science and Technology, Harbin Medical University Sep 2014 – Jul 2019

- Identified dysregulated sub-pathways in cancers by integrating gene expression, CNV, and DNA methylation data (co-author, *J Cell Mol Med.*, 2018).

PUBLICATIONS ¹

Summary: 11 peer-reviewed publications (7 as first / co-first author); 1 first-author preprint; 1 manuscript under review; 3 in preparation.

Long-read multi-omics profiling reveals structural variant heterogeneity and therapeutic vulnerability in acute lymphoblastic leukemia. (In preparation)

Li Z[†], Wang X[†], Jiang M[†], Zhuo Z[†], Tu Y[†], Wu D, Wang X[✉], Zhang H[✉], Shao J[✉]. **Mitochondrial ROS-driven HMGB1 translocation promotes autophagy and immune evasion in *MEF2D* fusion-positive B-ALL.** (In preparation)

Xue Q[†], Zhang M^{†✉}, Mo Y[†], Jiao B[†], Liu X[†], Jiang M, Zhou Y, Tan Y, Li H, Zhang J, Zhang Q, Li Y, Li J, Ma X, Jing DH, Mi JQ, Wang J, Chen Z, Shen SH[✉], Chen SJ[✉]. **Molecular mechanisms of unique therapeutic potential of CUDC-907 for *MEF2D* fusion-driven BCP-ALL.** *Signal Transduct Target Ther.* 2025 Jul 23;10(1):230. [10.1038/s41392-025-02310-y](https://doi.org/10.1038/s41392-025-02310-y)

Jiang M, Jing D, Wong JWH[✉]. **gghic: A Versatile R Package for Exploring and Visualizing 3D Genome Organization.** *arXiv.* 2024 Dec. [10.48550/arXiv.2412.03005](https://doi.org/10.48550/arXiv.2412.03005) (In preparation for submission to a peer-reviewed journal)

Ye J[†], Wang F[†], Zhou Z[†], Jiang M[†], You Y, Su Y, Hu Q, Chi H, Liu H, Cheng X, Jia J, Shi H, Yang C, Huang J[✉], Teng J[✉], Hu Q[✉], Sun Y[✉]. **Itaconate promoted monocyte migration by increasing chemokine CCL12 expression in autoinflammatory cytokine storm.** *Clin Immunol.* (Under review)

Li Z[†], Jiang M[†], Wang J[†], Zhuo Z[†], Zhang S, Tan Y, Hu W[✉], Zhang H[✉], Meng G[✉]. **Transcription factor 12-mediated self-feedback regulatory mechanism is required in *DUX4* fusion leukaemia.** *Clin Transl Med.* 2023 Dec;13(12):e1514. [10.1002/ctm2.1514](https://doi.org/10.1002/ctm2.1514).

Yin H[†], Wang J[†], Tan Y[†], Jiang M[†], Zhang H[✉], Meng G[✉]. **Transcription factor abnormalities in B-ALL leukemogenesis and treatment.**[#] *Trends Cancer.* 2023 Oct;9(10):855-870. [10.1016/j.trecan.2023.06.004](https://doi.org/10.1016/j.trecan.2023.06.004).

Wu W[†], Tan Y[†], Yin H[†], Jiang M[†], Jiang Y, Ma X, Yin T, Li Y[✉], Zhang H[✉], Cai X[✉], Meng G[✉]. **Phase separation is required for PML nuclear body biogenesis and function.** *FASEB J.* 2023 Jun;37(6):e22986. [10.1096/fj.202300216R](https://doi.org/10.1096/fj.202300216R).

Jiang M^{†✉}, Zhang S[†], Yin H[†], Zhuo Z[†], Meng G[✉]. **A comprehensive benchmarking of differential alternative splicing tools at the event level.** *Brief Bioinform.* 2023 May 19;24(3):bbad121. [10.1093/bib/bbad121](https://doi.org/10.1093/bib/bbad121).

Lu G[†], Ling Y[†], Jiang M[†], Tan Y[†], Wei D[†], Jiang L, Yu S, Jiang F, Wang S, Dai Y, Wang J, Wu G, Zhang X, Meng G[✉], Wang S[✉], Liu F[✉], Fan X[✉], Chen S[✉]. **Primary assessment of the diversity of Omicron sublineages and the epidemiologic features of autumn/winter 2022 COVID-19 wave in Chinese mainland.** *Front Med.* 2023 Aug;17(4):758-767. [10.1007/s11684-022-0981-7](https://doi.org/10.1007/s11684-022-0981-7).

Jiang M[†], Yin H[†], Zhang S[†], Meng G[✉], Wu G[✉]. **Mathematical appraisal of SARS-CoV-2 Omicron epidemic outbreak in unprecedented Shanghai lockdown.** *Front Med (Lausanne).* 2022 Nov 8;9:1021560. [10.3389/fmed.2022.1021560](https://doi.org/10.3389/fmed.2022.1021560).

Zhang M[†], Zhang H[†], Li Z[†], Bai L[†], Wang Q, Li J, Jiang M, Xue Q, Cheng N, Zhang W, Mao D, Chen Z, Huang J, Meng G[✉], Chen Z[✉], Chen SJ[✉]. **Functional, structural, and molecular characterizations of the leukemogenic driver *MEF2D-HNRNPUL1* fusion.** *Blood.* 2022 Sep 22;140(12):1390-1407. [10.1182/blood.2022016241](https://doi.org/10.1182/blood.2022016241).

Zhang H[†], Cheng N[†], Li Z[†], Bai L[†], Fang C, Li Y, Zhang W, Dong X, Jiang M, Liang Y, Zhang S, Mi J, Zhu J, Zhang Y, Chen SJ, Zhao Y, Weng XQ, Hu W[✉], Chen Z[✉], Huang J[✉], Meng G[✉]. **DNA crosslinking and recombination-activating genes 1/2 (*RAG1/2*) are required for oncogenic splicing in acute lymphoblastic leukemia.** *Cancer Commun (Lond).* 2021 Nov;41(11):1116-1136. [10.1002/cac2.12234](https://doi.org/10.1002/cac2.12234).

Wu L[†], Jiang M[†], Yu P, Li J, Ouyang W, Feng C, Zhao WL[✉], Dai Y[✉], Huang J[✉]. **Single-Cell Transcriptome Analysis Identifies Ligand-Receptor Pairs Associated With BCP-ALL Prognosis.** *Front Oncol.* 2021 Mar 10;11:639013. [10.3389/fonc.2021.639013](https://doi.org/10.3389/fonc.2021.639013).

Han J^{†∞}, Liu S[†], Jiang Y[†], Xu C[†], Zheng B, [Jiang M](#), Yang H, Su F, Li C[∞], Zhang Y[∞]. **Inference of patient-specific subpathway activities reveals a functional signature associated with the prognosis of patients with breast cancer.** *J Cell Mol Med.* 2018 Sep;22(9):4304-4316. [10.1111/jcmm.13720](https://doi.org/10.1111/jcmm.13720).

POSTERS ²

[Jiang M](#), Jing D, Wong JWH. **gghic: a versatile R package for exploring and visualizing 3D genome organization.** [GIW XXXIV ISCB-Asia 2025](#). InnoWing Phase 2, G/F, Chow Yei Ching Building, HKU, Hong Kong. 2025 Dec 11. *Brief Bioinform.* 2025 Dec;26(Suppl 1):i28. [10.1093/bib/bbaf631.042](https://doi.org/10.1093/bib/bbaf631.042) | [PDF](#)

Ma X, [Jiang M](#), Wang J, Mi JQ, Wong JWH, Jing D. **Long-read profiling of structural variants reveals mechanisms of chemoresistance and prognostic heterogeneity in acute lymphoblastic leukemia.** *Blood.* 2025 Nov;146(Suppl 1):1585. [10.1182/blood-2025-1585](https://doi.org/10.1182/blood-2025-1585) | [PDF](#) (accepted; not presented in person)

[Jiang M](#), Ma X, Ho YY, Yang E, Jing D, Wong JWH. **Long and short multi-omics identify structural variants and three-dimensional changes in B-cell acute lymphoblastic leukemia.** [SBMS Research Postgraduate Symposium 2025](#). Multidisciplinary Lab 1, Room LG-09, Laboratory Block, Faculty of Medicine Building, Hong Kong. 2025 May 15. | [PDF](#)

CONFERENCE TALKS ²

[Jiang M](#). **Long-read multi-omics profiling reveals chemoresistance heterogeneity and therapeutic vulnerability in acute lymphoblastic leukemia.** SBMS Research Postgraduate Symposium 2026. Lecture Theatre 2, William M.W. Mong Block, Faculty of Medicine Building, Hong Kong. 2026 May 21.

SOFTWARE & TOOLS

- [gghic](#)
A versatile R package for exploring and visualizing 3D genome organization
- [snakes](#)
A collection of Snakemake pipelines for comprehensive omics data analysis
- [mjstp](#)
An automated setup utility for development tools on Linux servers
- [MountManager](#)
A macOS menu bar application for managing SSHFS mounts via oxfS
- [Caffeinate](#)
A macOS menu bar app to keep your Mac awake on demand
- [RSWP](#)
RNA sequencing workflow by Python
- [ASP](#)
An alternative splicing package in R



TEACHING

Teaching Assistant / Demonstrator, School of Biomedical Sciences, The University of Hong Kong

Sep 2023 – present

- **Anatomy** — VR/AV practicals for MBBS students.
- **Biochemistry** — practicals on Southern blot analysis, subcloning, plasmid minipreps, and restriction mapping.
- **Examination invigilation**
- **Genome Informatics** — lectures and workshops on Linux/HPC, sequence alignment, variant calling, metagenomics, human/cancer genomics, cancer transcriptomics, and epigenetics.
- **Physiology** — practicals on nerve-muscle preparation, respiration, human reflexes, electrophysiology, EMG, and clinical examination.

- Extra tutorials on R for undergraduate students.

Full schedule available at mhjiang97.github.io/teaching.

PEER REVIEW & SERVICE

Journal peer review — invited reviewer:

- *Front Bioinform.* (2026)
- *Open Vet J.* (2026)
- *FASEB J.* (2024)

Journal peer review — co-reviewer (with supervisor):

- *Front Mol Biosci.* (2021, 2022)

AWARDS

First-class scholarship, Shanghai Jiao Tong University School of Medicine	2020, 2021
Outstanding graduate, Harbin Medical University	2019
Outstanding volunteer of The Belt and Road China–Russia Medical University Alliance	2018
Excellent student leader, Harbin Medical University	2016

OTHER ROLES & AFFILIATIONS

Member, [openbiox](#) open-source bioinformatics community



Director of Sports Department, Student Union, Shanghai Jiao Tong University School of Medicine

2019 – 2020

Director of Publicity Department, Student Union, Harbin Medical University

2014 – 2016

SKILLS

- **Programming languages:** proficient in R, Python, Shell; familiar with Perl and Java
- **HPC & workflow:** Linux, Slurm, PBS, Snakemake, Nextflow, Conda, Docker, Apptainer, Git/GitHub, Pixi, uv
- **Bioinformatics:** WGS, ChIP-seq, CUT&Tag, scRNA-seq, RNA-seq, CyTOF, 3D genome, long-read sequencing, variant calling, alternative splicing, cell–cell communication inference
- **Visualization & design:** ggplot2, Adobe Illustrator; familiar with Adobe Photoshop
- **Languages:** Mandarin (native), English (IELTS Academic 7.0; L 8.0 / R 7.0 / W 7.0 / S 6.0)

1. Reverse chronological order | [†] These authors contributed equally | [∞] Corresponding authors | [#] Review articles [↗](#)

2. Reverse chronological order [↗](#)